

MOLECULAR MECHANISMS OF PROTEIN-ASSISTED PASSIVE PERMEATION

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Keywords: permeation, drug, transmembrane proteins, molecular dynamics

ABSTRACT

Traditional modeling of passive permeation largely ignored the presence of transmembrane peptides, relegating them strictly to active transport roles. However, given the high protein density of biological membranes, permeants inevitably interact with transmembrane domains before entering the hydrophobic core. We investigated protein-assistance in the passive permeation of xenobiotics. Using biased all-atom molecular dynamics (MD) simulations¹, we evaluated the permeation of acetylsalicylic acid alongside four single-pass transmembrane domains (KALP15, KALP23, GFRAL, SEMA4D).

Our results reveal that the permeant translocates significantly faster along peptide surfaces than through purely lipidic environments. More importantly, the molecular mechanisms driving this enhancement are not protein-specific, hinting at a generalized molecular effect. By demonstrating that transmembrane domains can facilitate passive diffusion, this work opens the door to question current permeation paradigms and introduces a new parameter to consider for future pharmacokinetic modeling and drug development.

REFERENCES

- 1 Benmameri, M.; Chantemargue, B.; Humeau, A.; Trouillas, P.; Fabre, G. MemCross: Accelerated Weight Histogram Method to Assess Membrane Permeability. *Biochimica et Biophysica Acta (BBA) - Biomembranes* 2023, 1865 (3), 184120. <https://doi.org/10.1016/j.bbamem.2023.184120>.